MLX-FINETUNE-TEST — VISUAL PIPELINE

From raw dataset to a served, authenticated model

data prep fine-tuning serving

1 · DATA

SOURCE

Kaggle: Phishing Email Detection

18,650 emails · Email Text + Email Type

`subhajournal/phishingemails`

CLEAN

prepare_data.py

drop empty + 1,068 duplicates, cap length at 1,000 chars

`python scripts/prepare_data.py`

SPLIT

Stratified train/valid

9,356 kept → 8,420 train / 936 valid, ratio preserved

`train.jsonl · valid.jsonl`

2 · FINE-TUNING

BASE MODEL

Qwen3-4B-abliterated

4-bit MLX, 2.26GB, frozen during training

`mlx-community/Josiefied-Qwen3-4B`

TRAIN

LoRA fine-tune

1000 iters, batch 4 — only 0.18% of params trained

`mlx_lm.lora --train`

VERIFY

Checkpoint check

lowest val-loss checkpoint predicted one class only — picked by real output, not loss

`scripts/evaluate.py → 6/6`

3 · SERVING

FUSE

Merge adapter → fp16

LoRA weights folded into the base model

`mlx_lm.fuse --dequantize`

CONVERT

→ GGUF, then quantize

Q4_K_M: 8GB → 2.4GB

`convert_hf_to_gguf.py · llama-quantize`

SERVE

llama-server

OpenAI-compatible API, key required, CORS → localhost

`POST /v1/chat/completions`

`run_pipeline.py --serve` orchestrates the last 3 nodes end-to-end

all stages verified against real output